IMPERIAL COLLEGE LONDON

DEPARTMENT OF ELECTRICAL AND ELECTRONIC ENGINEERING
EXAMINATIONS

MSc and EEE/EIE PART IV: MEng and ACGI

SYSTEMS IDENTIFICATION AND LEARNING

Monday, 11 May 2026, 10:00 AM

Time allowed: 3:00 hours

There are FOUR questions on this paper.

Answer ALL questions.

All questions carry equal marks.

Any special instructions for invigilators and information for candidates are on page 1.

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Second Marker(s) : F. Teng

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SYSTEMS IDENTIFICATION AND LEARNING

1. Consider the stochastic process $v(\cdot)$ generated by the ARMA model

$$v(t) = -\frac{1}{3}v(t-1) + e(t) + ce(t-1)\,. \tag{1.1}$$

In Model (1.1), $e(\cdot) \sim WN(0,1)$ and c denotes an *unknown* parameter such that the ARMA Model (1.1) is in *spectral canonical form*.

a) The stochastic process $v(\cdot)$ is stationary. Why? Justify your answer.

[3 Marks]

b) The so-called normalized covariance of the stochastic process $v(\cdot)$ is given by

$$\rho(\tau) = \frac{\mathbb{E}[v(t)v(t-\tau)]}{\text{var}[v(t)]}\,. \tag{1.2}$$

For $\tau = 1$, $\rho(\tau)$ takes on the value

$$\rho(1) = -1/3\,.$$

Determine the value of the parameter c.

[3 Marks]

c) For the value of c determined in your answer to Question 1b), write the difference equation representing the optimal **two-step ahead** predictor $\hat{v}(t+2|t)$ of $v(t)$.

[5 Marks]

d) Now, assume that, for $\tau = 1$, the normalized covariance $\rho(\tau)$ given in (1.2) takes on the value

$$\rho(1) = 3/10\,.$$

Determine the value of the parameter c.

[6 Marks]

e) For the value of c determined in your answer to Question 1d), write the difference equation representing the optimal **one-step ahead** predictor of $v(t)$.

[3 Marks]

2. Consider the block scheme shown in Fig. 2.1 representing a linear discrete-time dynamical system generating a stochastic process $y(\cdot)$.

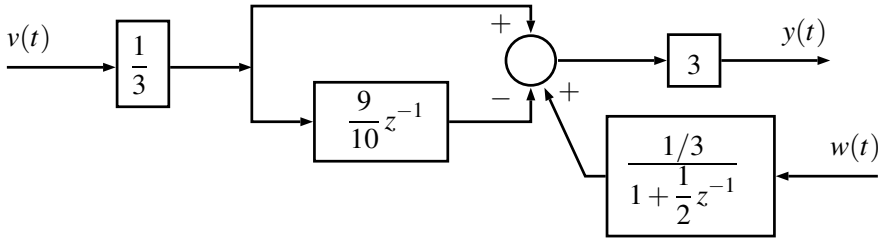


Figure 2.1 Dynamic system generating the stochastic process $y(\cdot)$.

The inputs $v(\cdot) \sim WN(0,1)$ and $w(\cdot) \sim WN(0,1)$ are mutually uncorrelated white processes.

a) Write the difference equation expressing the dynamic model of the stochastic process $y(\cdot)$.

[4 Marks]

b) Determine the expected value $\mathbb{E}[y(t)]$ of the stochastic process $y(\cdot)$.

[4 Marks]

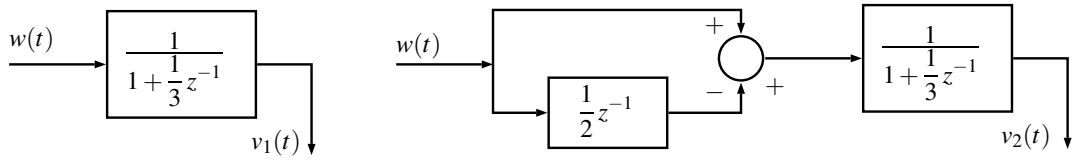
c) Determine the variance $\text{var}[y(t)]$ of the stochastic process $y(\cdot)$.

[4 Marks]

d) Determine the spectrum $\Gamma_y(\omega)$ of the process $y(\cdot)$ and sketch its behaviour in the angular frequency domain $\omega \in [-\pi, \pi]$.

[8 Marks]

3. Consider two different stochastic processes: **Process 1** denoted by $v_1(\cdot)$ and **Process 2** denoted by $v_2(\cdot)$ generated as illustrated in Fig. 3.1.



Process 1 **Process 2**

Figure 3.1 Two stochastic data-generation mechanisms.

Moreover, consider the family of AR(1) stochastic models

$$\mathscr{M}(\theta): \quad y(t) = ay(t-1) + w(t), \quad \theta := a$$

where, with reference to **Process 1** in Fig. 3.1, we set $y(t) = v_1(t)$, whereas with reference to **Process 2** in Fig. 3.1, we set $y(t) = v_2(t)$.

In both **Process 1** and **Process 2**, consider an MEP identification procedure that minimises the prediction error in which $\widehat{a}(N)$ is the least squares estimate of the parameter a based on N measurements $\{y(1), y(2), \ldots, y(N)\}$.

a) **Process 1.** The stochastic process is $y(\cdot) = v_1(\cdot)$ with $w(\cdot) \sim WN(0,1)$. Determine the value $\bar{a}_1$ to which the estimate $\widehat{a}(N)$ approaches for large values of N (that is, $\bar{a}_1 = \lim_{N\to\infty} \widehat{a}(N)$, a.s.).

[6 Marks]

b) **Process 2.** The stochastic process is $y(\cdot) = v_2(\cdot)$ with $w(\cdot) \sim WN(0,1)$. Determine the value $\bar{a}_2$ to which the estimate $\widehat{a}(N)$ approaches for large values of N (that is, $\bar{a}_2 = \lim_{N\to\infty} \widehat{a}(N)$, a.s.).

[7 Marks]

c) With reference to **Process 1** of Question 4-a) and **Process 2** of Question 4-b), compute and compare the variances of the prediction errors

$$\varepsilon_1(t) = \text{var}[v_1(t) - \bar{a}_1 v_1(t-1)] \quad \text{and} \quad \varepsilon_2(t) = \text{var}[v_2(t) - \bar{a}_2 v_2(t-1)]$$

Comment on your findings.

[4 Marks]

d) Consider a different variance for the process $w(\cdot)$, namely $w(\cdot) \sim WN(0,3)$. State whether or not the values of $\bar{a}_1$ and $\bar{a}_2$ computed in the answers to Questions 4-a) and 4-b) are different because of the different value taken on by the variance of $w(\cdot)$. Justify your answer.

[3 Marks]

4. Consider the electrical circuit depicted in Fig. 4.1.

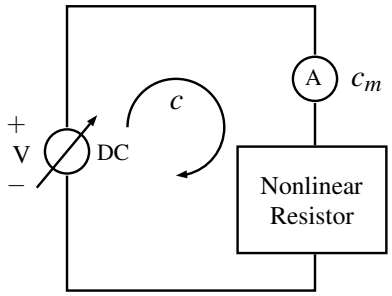


Figure 4.1 Electrical circuit with a nonlinear resistive load and a DC power supply.

A current meter is available providing a measurement c_m of the current c flowing in the circuit. An experiment to **learn** the function characterising the nonlinear resistor is set up. A set of M given values of time-samples $t(j)$, $j = 1,\ldots,M$ are selected. Correspondingly, a set $V(j)$, $j = 1,\ldots,M$ of different values of the voltage of the power supply are selected and a set of M respective measurements $c_m(j)$, $j = 1,\ldots,M$ of the current are recorded by the current meter:

$$c_m(j) = c(j) + \xi(j),\ j = 1,\ldots,M$$

where $c(j)$ is the true current flowing in the resistor at time $t(j)$ and ξ is a zero-mean white process $\xi(\cdot) \sim WN(0,\lambda^2)$.

a) Consider the following two *parametric* models $\mathscr{M}_i$, $i = 1,2$ of the unknown current $c(t)$ of the form

$$\mathscr{M}_1 :\ \widehat{c}[V(j),\alpha] = \alpha,\ j = 1,\ldots,M \qquad \text{(Constant Model)}$$
$$\mathscr{M}_2 :\ \widehat{c}[V(j),\alpha,\beta] = \alpha + \beta \cdot \arctan[V(j)],\ j = 1,\ldots,M \qquad \text{(Nonlinear Model)}$$

where α,β,γ are scalar parameters to be determined using the available set $V(j)$, $j = 1,\ldots,M$ of values of the voltage of the power supply and the set of current measurements $c_m(j)$, $j = 1,\ldots,M$.

i) For *parametric* model $\mathscr{M}_1$, devise a least-squares method the solution of which provides the optimal (in the least-squares sense) value of the parameter α and determine its solution in analytical form.

[5 Marks]

ii) For *parametric* model $\mathscr{M}_2$, devise a least-squares method the solution of which provides the optimal (in the least-squares sense) value of the parameters α,β and determine its solution in analytical form.

[7 Marks]

b) Consider the following set Θ of $M = 5$ values of voltage $V(j)$, $j = 1,\ldots,M$ and respective current measurements $c_m(j)$, $j = 1,\ldots,M$:

$$\Theta = \{(1,1.34),(2,1.58),(3,1.75),(4,1.79),(5,1.9)\}$$

Using the general solution obtained in your answer to Question 4-a-i) and 4-a-ii), compute the optimal parameter α for model $\mathscr{M}_1$ and optimal parameters α,β for model $\mathscr{M}_2$. Sketch the plot of the characteristic of the resistor of the obtained approximate models and also plot on the same diagram the 5 points of the set Θ. Discuss your findings.

i) Correct answer for $\mathscr{M}_1$ gets [3 Marks]

ii) Correct answer for $\mathscr{M}_2$ gets [5 Marks]